



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review Updated Article for IEEE Access Access-2024-40922

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Tue, Oct 22, 2024 at 2:03 AM

Reply-To: arianna.dulizia@irpps.cnr.it

To: khare.pranav@gmail.com

22-Oct-2024

Dear Mr. Khare:

Recently you reviewed an article submitted to IEEE Access entitled, "An Innovative Solution to Design Problems: Applying the Chain-of-Thought Technique to Integrate LLM-based Agents with Concept Generation Methods". Please note that this article was rejected, and the authors were asked to update their article based on the reviewers' concerns.

The authors have resubmitted their revised article and supplied a "list of updates" that highlights all edits made, as well as their responses to the previous reviewers' concerns.

We would appreciate if you could take another look at this improved version to determine if your concerns have been addressed, and to ensure that you are satisfied with the updates.

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Dr. Arianna Dulizia
IEEE Access Associate Editor
arianna.dulizia@irpps.cnr.it

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: An Innovative Solution to Design Problems: Applying the Chain-of-Thought Technique to Integrate LLM-based Agents with Concept Generation Methods

AUTHORS: Ge, Shijun; Sun, Yuanbo; Cui, Yin; Wei, Dapeng

ABSTRACT: To enhance the application capabilities of large language models (LLMs) in conceptual design, this study explores how to achieve deep integration between LLM-based agents and concept generation methods using the chain-

of-thought (CoT) technique and evaluates its feasibility. Using GPT-4 as a case study, we designed two agents: IntelliStorm (based on the unstructured brainstorming method) and EvoluTRIZ (based on the structured TRIZ method). Thirty participants were recruited, and through two experimental phases spaced one month apart, a comparative analysis of the effects of collaboration groups (human-agent vs. human-human) and concept generation methods (brainstorming vs. TRIZ) on participants' physiological activation and creative thinking performance were conducted. The results show that the involvement of LLM-based agents can effectively reduce participants' electrodermal activity (EDA) response levels, indicating a reduction in cognitive load. Moreover, these agents preserve the distinct physiological response patterns and performance advantages of the different concept generation methods. For example, IntelliStorm, like brainstorming, evokes stronger responses to information stimuli, demonstrating superior thinking fluency; EvoluTRIZ, like the TRIZ, exhibits a higher frequency of information responses, showcasing enhanced thinking elaboration. However, originality tends to favor human-human collaboration. The findings confirm that integrating LLMs with traditional concept generation methods is an effective strategy made possible by combining CoT and retrieval-augmented generation (RAG) technologies. In the future, LLM-based agents are expected to achieve broader application in the design field by incorporating additional concept generation methods.

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